



NextEdge
Technologies

OID migration Proposal

**Technical Proposal
To Credit Agricole**

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Company Details

HQ

1, Mostafa El nahas, Nasr City, Cairo Egypt

info@nxtedgetechnologies.com

Contacts

CEO	Mostafa El kady
Mobile	+20 1005298759
E-mail	m.elkady@nxtedgetechnologies.com
Account Manager	Amr EL kady
Mobile	+20 1003910014
E-mail	A.elkady@nxtedgetechnologies.com

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1. Executive Summary

NxtEdge has the pleasure to seize the opportunity to submit this proposal in response to Credit Agricole Egypt request for Migration the Current retail digital channel directory server to OpenLDAP.

Based on our Experience in the Banking industry and experience in directory service implementation the below offer is covering all aspect of service considering CBE related regulation and criticality of service.

We have mapped CAE requirements and existing solution capabilities to the new directory service capabilities.

This document presents the technical offer for the implementation of an enterprise-grade OpenLDAP directory service using a multi-master MirrorMode replication model. The solution ensures high availability, scalability, and disaster recovery readiness, while providing a secure and centralized identity store for customer authentication and application integration.

As a result of NXTEDGE's in depth understanding of CAE business needs and its intimate familiarity with the required solution, we believe that NXTEDGE is primly positioned to offer CAE effective and efficient delivery and deployment of the required solution for an optimum duration which is estimated to be 3 calendar months.

We trust that this document presented to NXTEdge will represent the best fitting solution to comply with CAE requirements and that it meets your satisfaction. We are hopeful that you find this document satisfactorily to serve its purpose and is in line with CAE business requirements, while at the same time providing flexibility to change as the business dictates.

2. Scope of work

The following are full details about the suggested components,

2.1.1 OpenLdap implementation

The scope of this engagement includes the following:

1. Analysis Phase

- Assess current authentication and identity infrastructure.
- Identify business requirements (availability, performance, compliance).
- Review existing applications and systems that will integrate with LDAP.
- Gather schema, attribute, and security requirements.
- Define user population size, growth projections, and query patterns.
- Document integration and reporting needs (e.g., SIEM, monitoring, audit).

2. Design & Architecture

- Propose Directory Information Tree (DIT) structure and schema extensions.
- Define indexes, ACLs, and password policies.
- Develop high-availability architecture with **MirrorMode Multi Master replication** across main data center
- Plan for optional read-only replicas.
- Identify backup, monitoring, and failover strategy.

3. Implementation

- Install and configure OpenLDAP servers (2 masters: main +2 read only replica) in test and production environment.
- Configure LDAPS.
- Configure replication using **delta-sync REPL** in MirrorMode.
- Implement two read-only replicas per environment.
- Harden security (TLS/SSL, ACLs, service accounts).

4. Integration

- Create service account for each application to authenticate via LDAP.
- Validate schema mapping with applications.

5. Testing

- Functional testing (binds, searches, writes).
- Failover and recovery testing (promotion of DR site).
- Confirm DR recovery using VM replicas.
- Tune the system based on performance testing results.

6. Knowledge Transfer & Documentation

- Deliver system architecture and configuration documentation.
- Conduct 1 day admin knowledge and handover session.

3. Technical Solution

3.1 Architecture Overview

- **Main Site (Active-Active)**
 - **Master A** – write or full read/write
 - **Master B** – write or full read/write.
 - **Read replica A1**-Read
 - **Read replica B1**-Read
 - Bi-directional delta-sync repl ensures data consistency.
 - Two Load balancer (one for read and one for write) provides a single LDAP endpoint for applications.
- **DR Site**
 - Bank's **VM replication** maintains a copy of the LDAP environment.
 - In a DR event, VM replicas are activated and become authoritative.
 - No native LDAP replication between sites (simplifies ops, leverages bank's infra).

3.2 Security

- Enforce TLS/SSL for all LDAP traffic needed.
- Strong ACLs to enforce least-privilege access.
- Password policies and lockout configured.

3.3 High Availability

- **Multi-master** with mirror mode at the main site which guarantee resilience against node failure. A description for the multi-master mirror mode is described below
 - MirrorMode is a special form of multi-master replication in OpenLDAP.
 - In this model, two servers (masters) is configured with bi-directional replication (each replicate to the other).
 - Technically, both servers can accept read and write operations.
 - However, the system is designed so that only one master is active for writes at a time.
 - The other master acts as a hot standby (read-only in practice), maintaining an up-to-date copy through replication.
 - If the active master fails, the standby can be quickly promoted to writer, ensuring continuity.

Why MirrorMode is Chosen

- Consistency > concurrency: In regulated industries such as banking, having a single authoritative writer is more important than allowing writes on multiple nodes simultaneously.

- Operational simplicity: MirrorMode provides the HA benefits of multi-master without the conflict-resolution complexity of full active-active replication.
- Simplicity for future data warehouse and CDC integration.
- Enterprise fit: It strikes the right balance between availability, reliability, and manageability, making it a standard recommendation for enterprise directory deployments.

- VM replication to DR → site-level resilience with banking infrastructure.
- Load balancer ensures seamless failover between nodes.

3.4 Hardware requirements

The following hardware specifications are recommended to support the deployment of OpenLDAP in a **2-node multi-master architecture with 2 read replicas**. These requirements are based on best practices for enterprise directory services, ensuring performance, scalability, and high availability.

3.4.1 Server Roles

- **Master Nodes (2 servers, Main Site)**

- Handle all read and write or write only operations to be defined in the design phase.
- Provide replication to each other and to replicas.
- Serve as the authoritative directory source.

- **Read Replica Nodes (2 servers, Main Site)**

- Handle read-only traffic (authentication, searches).
- Offload queries from master nodes.

- **DR Site**

- Provided by the bank's VM replication infrastructure.
- Mirrors the main site environment for disaster recovery.

3.4.2 Recommended Specifications

CPU

- Minimum: 8 vCPU (2.5+ GHz)
- Recommended: 12–16 vCPU for masters, 8 vCPU for replicas

Memory (RAM)

- Minimum: 16 GB per node
- Recommended: 32 GB (to ensure LMDB caches full working set in memory for high-performance lookups)

Storage

- Type: Enterprise-grade SSD/NVMe
- Capacity:
 - Masters: 500 GB (data + accesslog + growth buffer)
 - Replicas: 300 GB
- Enable SAN-backed redundancy

Networking

- 10 Gbps uplink for high-volume environments

Operating System

- RHEL 8/9

3.4.3 Additional Considerations

- NTP Synchronization: Required across all nodes to ensure replication accuracy (contextCSN).
- TLS/SSL Certificates: Issued by customer PKI/CA for LDAPS traffic.
- Backup Storage: At least 2× directory size available on separate backup storage for mdb_copy and slapcat exports.
- High Availability: Place Master A and Master B on separate physical hosts/racks to avoid single points of failure.

3.5 Warranty

- The implemented OpenLDAP solution will be covered by a three (3) month warranty period starting from the official date of project acceptance and handover.
- During this warranty period, the following will apply:

3.5.1 Scope of Warranty

- Correction of any defects, errors, or misconfigurations in the implemented solution that are directly related to the scope of work defined in this proposal.
- Support for troubleshooting issues arising from the deployed OpenLDAP components, replication setup, and integrated overlays (e.g., lastbind, ppolicy, accesslog).
- Assistance in resolving failover or replication inconsistencies between the main and standby/replica nodes.

3.5.2 Response and Resolution

- Issues reported within the warranty period will be addressed according to agreed service levels (response time within 2 business hours).
- Remote support will be provided primarily.

3.5.3 Exclusions

- Changes or incidents caused by factors outside the project scope (e.g., unsupported modifications, operating system misconfigurations, hardware/network failures, or third-party software issues) are not covered under the warranty.
- New features, schema changes, or integrations requested after project acceptance will be treated as change requests.
- At the end of the warranty period, the customer may choose to enter into a support and maintenance agreement to continue receiving ongoing assistance, updates, and enhancements.

3.6 Prerequisites

- Four Linux servers VMs per environment with Linux Redhat installed
- Two Load balancer per environment, one is dedicated for read access and the other to write access to achieve high-availability client endpoint.
- TLS certificates from internal PKI/CA.
- Bank's VM replication mechanism in place for DR.
- NTP synchronization across all nodes.
- Remote Access to the environment.

3.7 Assumptions and out of scope

- Remote access to the environment will be provided by CAE
- CAE to provides VM and operating system installation and Hardening.
- Configure load balancing (VIP/DNS cutover) is out of scope.
- Performance and load testing is out of scope.
- CAE to provide VM replication setup for DR site.
- CAE to provide infrastructure, OS, and certificates.
- Failover at DR site is triggered and managed by CAE DR procedures however any requirement to apply such procedure will be supported.
- No direct LDAP replication between main and DR sites.
- The Migration strategy for data from the existing OID to the openLDAP to be defined during the design phase.
- Any document requiring approval by CAE to be approved within 3 business days.

3.8 Deliverables

- Fully functional 2-node multi-master OpenLDAP at the main site.
- DR environment aligned with bank's VM replication solution.
- Secure schema, DIT, ACL, and policy configuration.
- Documentation and operational runbooks.
- Knowledge transfer to customer admins.

4. TimeLine

The OpenLDAP implementation will follow a phased approach over a **3-month period**, ensuring quality delivery, stakeholder engagement, and risk mitigation.

4.1 Project Phases and Duration

Phase	Activities	Duration
1. Analysis (Weeks 1–2)	Requirements gathering, application inventory, schema/ACL/policy workshops, DR approach validation	2 weeks
2. Design (Weeks 3–4)	DIT structure, schema definition, replication model (2-node multi-master), indexes, ACLs, security model, architecture documentation, design approval	2 weeks
3.Implementation (Weeks 5–8)	Installation & configuration of 2 multi-master nodes, setup of 2 read replicas, replication (delta-syncrepl),	4 weeks

Phase	Activities	Duration
	overlays (lastbind, ppolicy, accesslog), TLS/SSL hardening	
4. Integration (Weeks 9–10)	Load balancer/DNS configuration, ACL validation, performance tuning Validating and testing the LDAP Testing the integration	2 weeks
5. Testing (Weeks 11–12)	Functional testing (bind, search, modify), replication & failover testing, DR restore simulation with VM replication	2 weeks
6. go live ,Handover & Closure (Week 12)	Documentation, knowledge transfer sessions, project acceptance and transition to warranty	1 week (overlaps with testing completion)

4.2 Milestones

- **End of Week 2** – Analysis completed; requirements validated
- **End of Week 4** – Design finalized and signed off
- **End of Week 8** – Multi-master + replicas implemented
- **End of Week 10** – integration completed
- **End of Week 12** – Testing successful, Go live ,documentation delivered, knowledge transfer completed

5. Optional Items

5.1 (Optional) APIs Integration with OpenLDAP

5.1.1 Modify the existing APIs

5.1.1.1 Objective & Scope

Refactor the belowlisted APIs to authenticate against and update identities in the new OpenLDAP directory (2-node multi-master at Main site; DR via bank VM replication). Scope includes:

- Replace legacy OID and OAM calls with **LDAP bind/search/modify** operations.
- Preserving existing business flows only shifting identity source and password ops to LDAP.

Business flow	API Endpoint
Change Password	changePassword
Forget password	sendPasswordOTP
	resetUserPass
Self registration	verifyCardCvv
	verifyCardPin
	verifyCustomerTpin
	validateUsernameAndPass
	getRegQuestions
	registerUser
	changeChallengeQuest
	changeSecurityQuest
Authenticateuser	authenticateUser
Getuserinfo	getUserInfo
	retrieveUserData
Session	addSession
	verifySession
	removeSession
Challenge Question	getChallengeQuestionForUser
	changeSecurityQuest
	changeChallengeQuest
	resetChallengeQuest
	sendChallengeQuestOTP
Change mail	changeEmail
Biometric	sendBiometricOTP
	validateBiometricOTP

Forget Username	sendUsernameOTP
	validateUsernameOtp

5.1.2 Create the following new APIs

Business flow	API Endpoint
Activate user(first time activation)	Activate user
Deactivate user	Deactivate user
reactivate user	Reactivate user
Delete User	Force Delete Existing user